



## Intuition Behind the Gradual Increase of Noise Variance in Diffusion Models

Asked 1 year, 2 months ago Modified 13 days ago Viewed 570 times

I've been studying diffusion models and came across the noise schedule, particularly how the noise variance  $\beta_t$  is adjusted over iterations. I've observed that  $\beta_t$  typically starts from a very small value during the initial steps and increases to a much larger value in the final steps. What is the underlying reason for this progression? Why is it important to begin with a low noise variance and end with a high one? I'd appreciate any intuitive explanations or references that shed light on this choice.

diffusion-models

Share Improve this question Follow

asked Sep 11, 2023 at 8:03

andy90  
111 1

I recommend reading this blog post from diffusion expert Sander Dieleman "[Noise schedules considered harmful](#)", which includes some very clear explanation  
– Dan Stowell Nov 10 at 22:05

### 2 Answers

Sorted by: Highest score (default)

mainly because it works, it makes learning the denoising process stable, because at each step the added noise is not too much, it makes it easy for the AI to figure out the changes that occurred the try and remove the added noise, plus the changes in its weight won't be large for each denoising step, am pretty sure you have been in a math class and have had to simplify some complex equation, imagine your teacher shows you the complex long equation, and then he just writes the simplified equation under it without showing the steps ? .... yeah, that will make it hard for you to learn anything, you can still figure it out because you are human, and your teacher already taught you about BODMAS and some rules of math but still it will be difficult, most diffusion are just that, diffusion models, they don't have human level reasoning etc.

Even for humans, when changes happen rapidly, it makes it hard to predict what actually happened. Imagine how hard it will be for you if you haven't learnt about BODMAS and stuff and you are exposed to math expression and the final solution? Gradual steps lead to better learning in humans, it works for ai too.

Yeah, so you begin with small steps, hopefully during the small steps the ai learns the fundamental rules of the operations, which is equivalent to a teacher showing each step because you are new to the material, then after you have some familiarity, teachers begin to skip steps and sometimes don't show steps at all.

Share Improve this answer Follow

answered Sep 11, 2023 at 10:44

Mubarak Salley  
1

Let me provide an intuitive explanation. During the forward diffusion process in the denoising diffusion model, the variance of the noise needs to be increased to **maintain a relatively constant Signal-to-Noise Ratio (SNR)**. This is probably because a stable SNR makes it easier for the network to learn denoising.

You can think of SNR as the ratio between the signal's variance and the variance of the added Gaussian noise. As we progressively add noise to the image, the SNR naturally decreases. To counteract this decrease and keep the SNR stable, we need to increase the variance of the noise added in the later stages of the process.

Share Improve this answer Follow

edited Jun 12 at 1:41

answered Jun 12 at 1:34

lxhfhut  
1 1